CNthesizer

User Guide

About CNthesizer

CNthesizer is an integrated AI-based platform developed to support the efficient exploration of compression ignition fuel design space.

Motivated by the continued importance of diesel engines in transport and industry, and the need to reduce emissions from conventional fossil fuels, the platform combines property prediction, mixture and biodiesel analysis, fuel screening, optimisation, and generative molecular design within a single workflow.

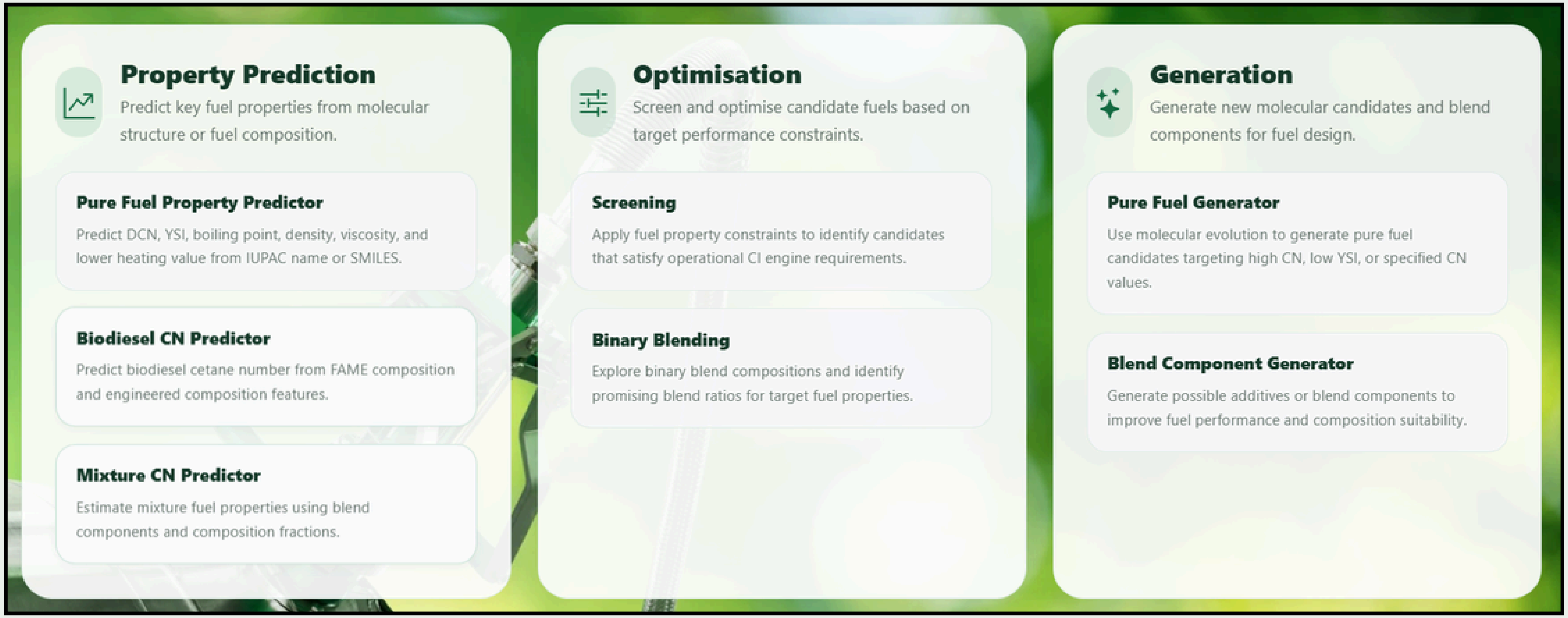
The platform focuses on key fuel properties that govern ignition behaviour, combustion performance, emissions potential, and practical fuel suitability. These include cetane number (CN), yield sooting index (YSI), boiling point, density, viscosity, and lower heating value (LHV).

Image: istockphoto.com/

Contents

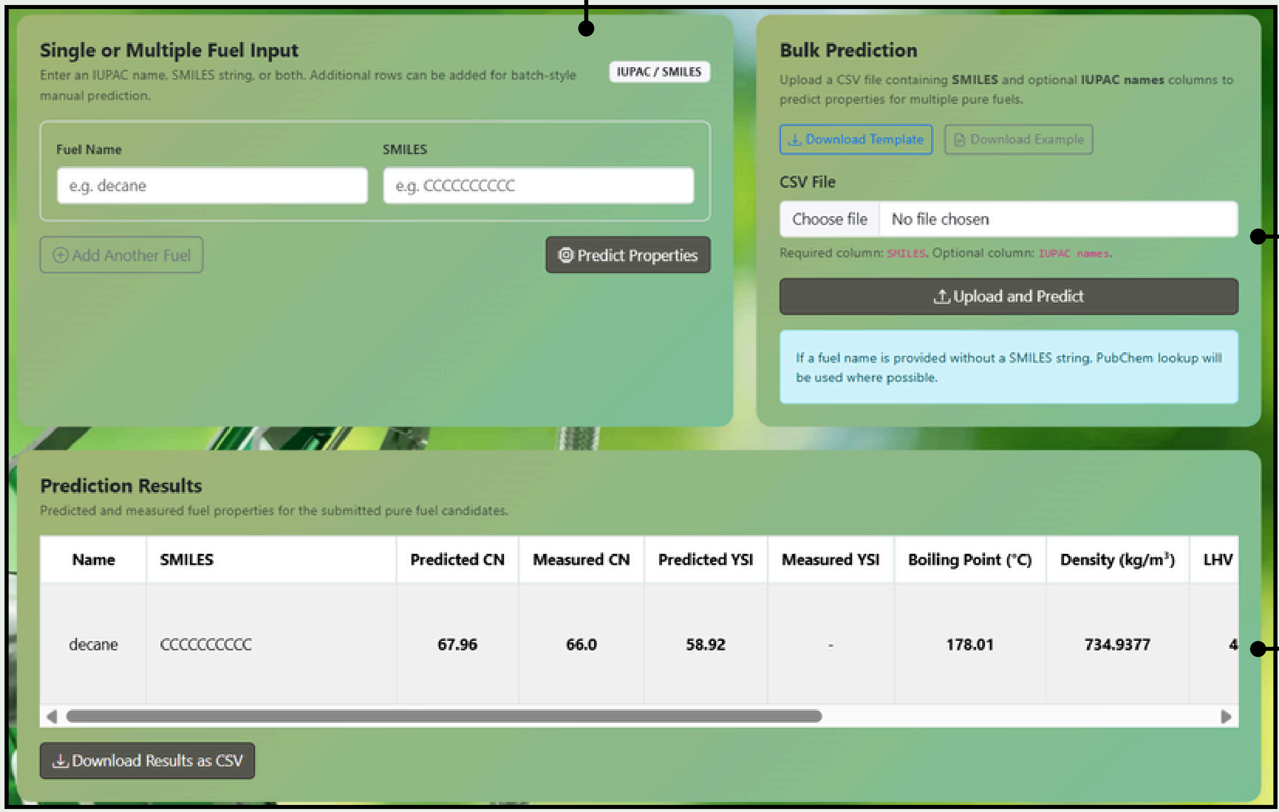
Tool Selection

Select the tool you require from the dashboard.



Pure Fuel Property Predictor

The tool gives a choice of two modes: manual entry and CSV upload.



Manual Entry
Enter one or more molecules using their IUPAC names or SMILES strings.

CSV Upload
Upload a CSV file containing a SMILES column and an optional IUPAC names column.

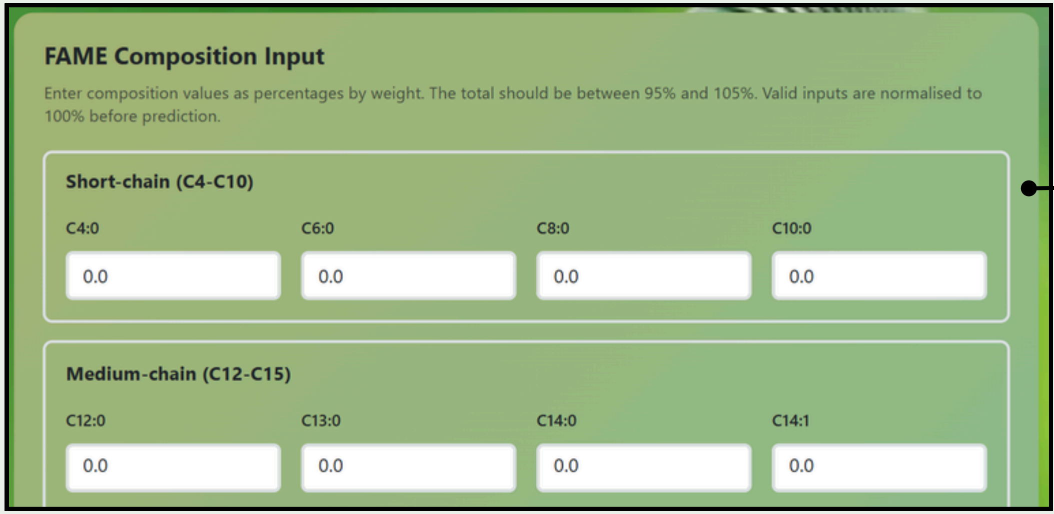
Template and **example** CSV files are provided to help you format your data correctly.

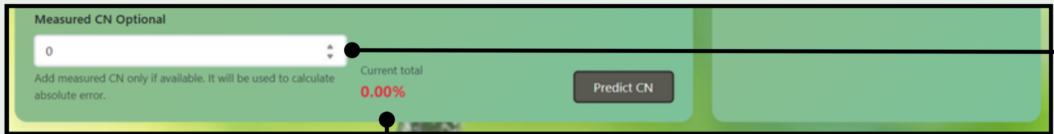
Results
Predicted properties are given for each molecule with measured CN and YSI where available. Any errors are flagged in the Status column. Results can be exported as a CSV file.

Biodiesel CN Predictor

The dropdown menu gives a choice of two modes: single entry and CSV upload.

Single Entry





Enter the weight percentage of each FAME component.

For valid inputs, the total composition must be **between 95% and 105%**, C18:3 **≤ 12 wt%**, and **C18:2 ≤ 70 wt%**. Invalid entries will be flagged automatically.

Enter the measured CN of the biodiesel sample to compare the predicted and experimental values.

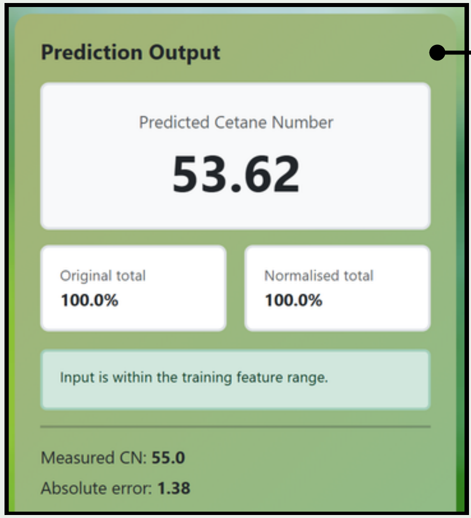
Monitor the total composition.

CSV Upload

Upload a CSV file containing the FAME composition of each biodiesel sample. Any missing FAME columns will automatically be assigned a value of zero.

Template and **example** CSV files are provided to help you format your data correctly.

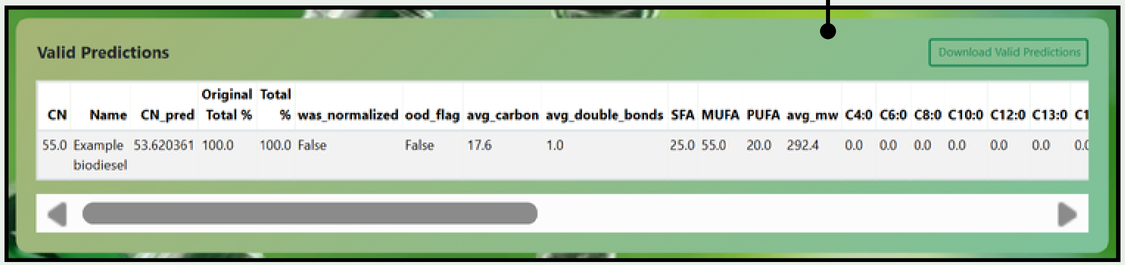
Results



The results summary displays the predicted CN and, where available, the absolute error relative to the measured CN.

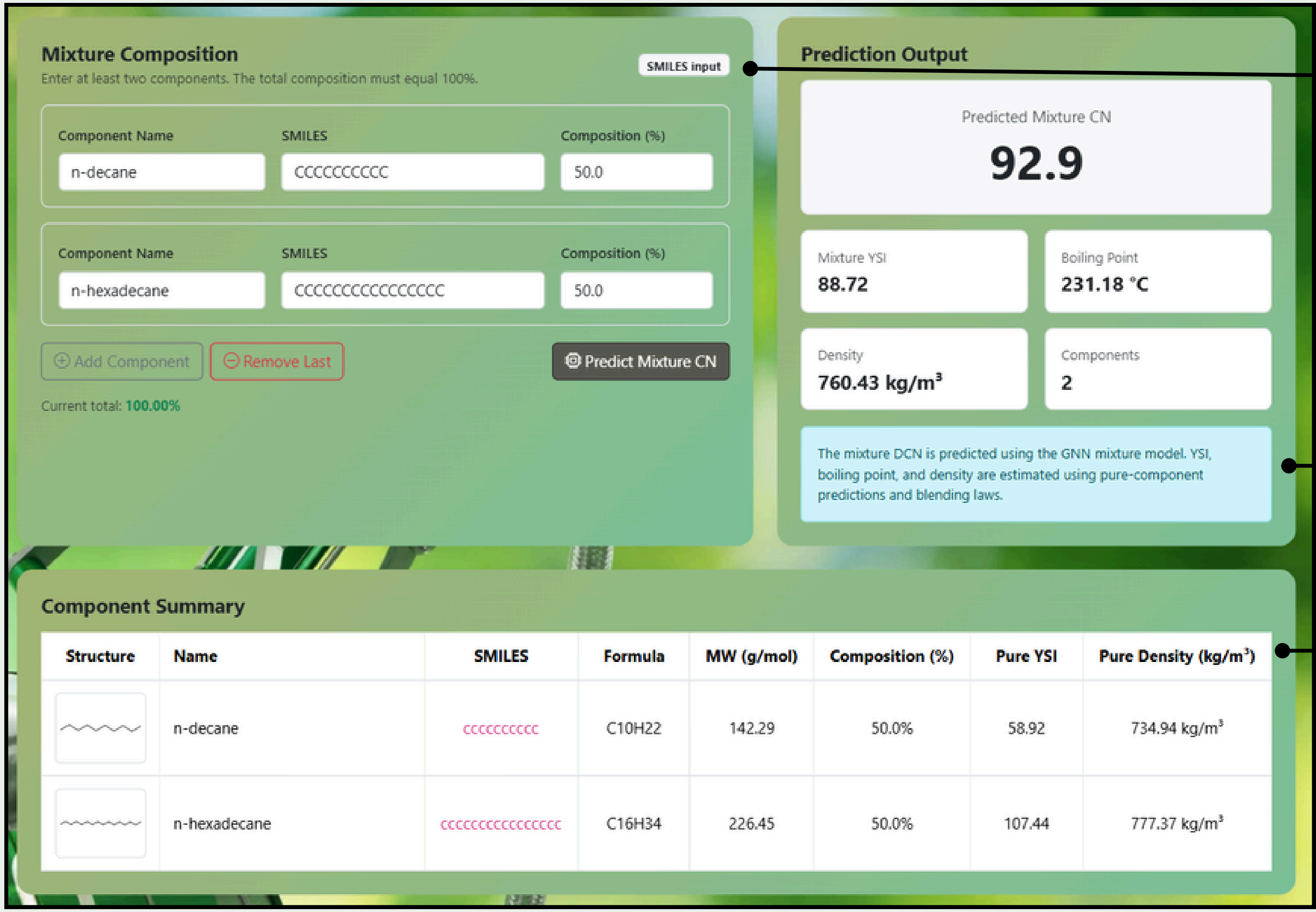
Inputs outside the model's training domain are flagged automatically.

The full results also include the engineered features and can be downloaded as a CSV file.



CN	Name	CN_pred	Original Total %	Total %	was_normalized	ood_flag	avg_carbon	avg_double_bonds	SFA	MUFA	PUFA	avg_mw	C4:0	C6:0	C8:0	C10:0	C12:0	C13:0	C1
55.0	Example biodiesel	53.620361	100.0	100.0	False	False	17.6	1.0	25.0	55.0	20.0	292.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Mixture CN Predictor



Input
Enter the IUPAC name or SMILES of each component together with its mole percentage.

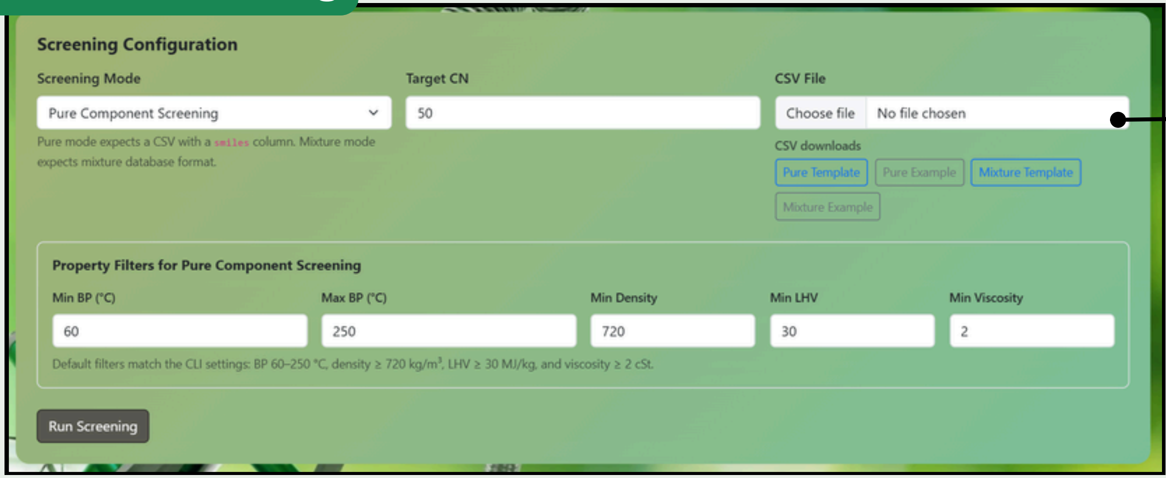
You may enter up to **6** components. Monitor the total composition at the bottom. It must be equal to **100%**.

Results
The output includes the predicted CN of the mixture and its estimated YSI and density. Predicted YSI and density for the individual components are also given.

Screening

The dropdown menu offers two modes: pure fuel screening and mixture screening.

Pure Fuel Screening



Upload a CSV file containing the SMILES of each candidate and enter a target CN. Adjust the property filters below if needed.

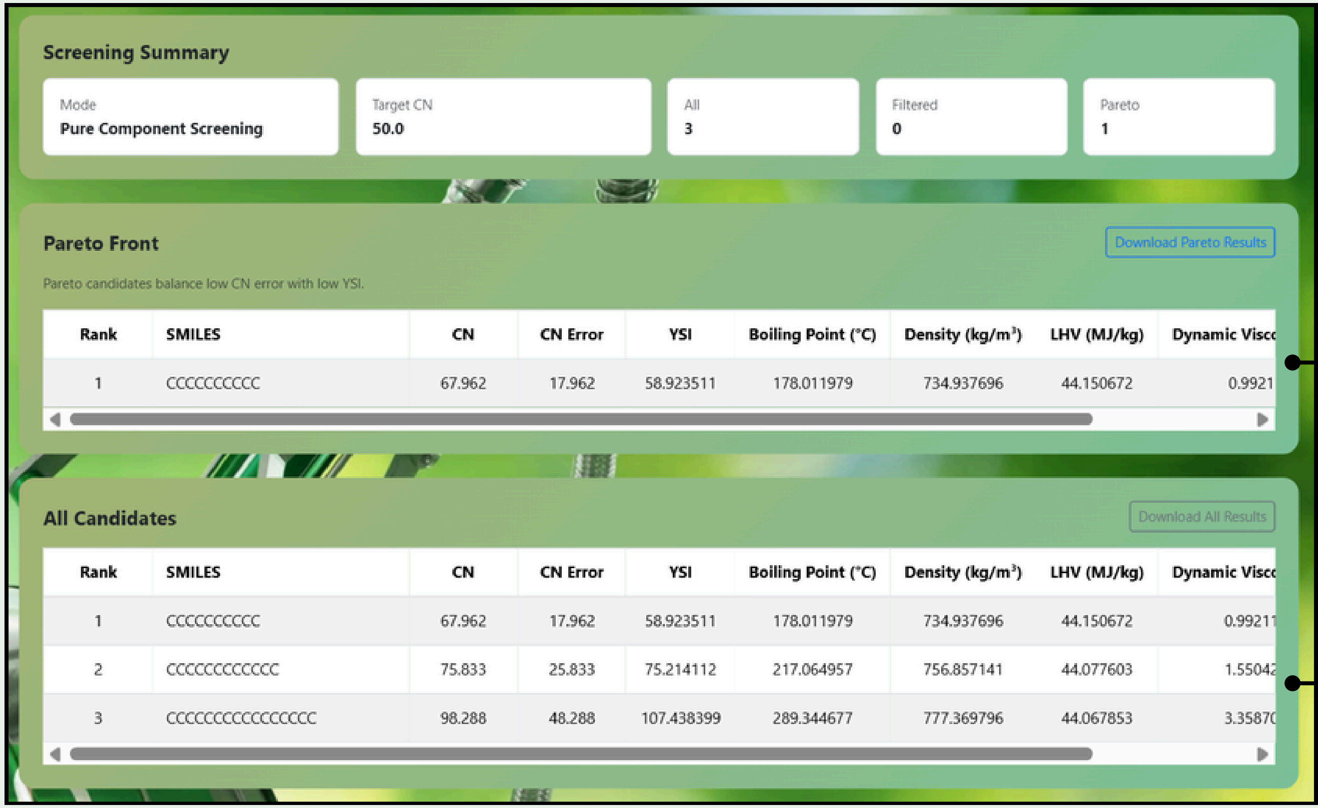
Template and **example** CSV files are provided to help you format your data correctly.

Mixture Screening

Upload a CSV file with component SMILES and mole fractions for each candidate and enter a target CN.

Template and **example** CSV files are provided to help you format your data correctly.

Results



Screening Summary

Mode	Target CN	All	Filtered	Pareto
Pure Component Screening	50.0	3	0	1

Pareto Front

Download Pareto Results

Pareto candidates balance low CN error with low YSI.

Rank	SMILES	CN	CN Error	YSI	Boiling Point (°C)	Density (kg/m³)	LHV (MJ/kg)	Dynamic Visc
1	CCCCCCCCCC	67.962	17.962	58.923511	178.011979	734.937696	44.150672	0.9921

All Candidates

Download All Results

Rank	SMILES	CN	CN Error	YSI	Boiling Point (°C)	Density (kg/m³)	LHV (MJ/kg)	Dynamic Visc
1	CCCCCCCCCC	67.962	17.962	58.923511	178.011979	734.937696	44.150672	0.99211
2	CCCCCCCCCCCC	75.833	25.833	75.214112	217.064957	756.857141	44.077603	1.55042
3	CCCCCCCCCCCCCCCC	98.288	48.288	107.438399	289.344677	777.369796	44.067853	3.35870

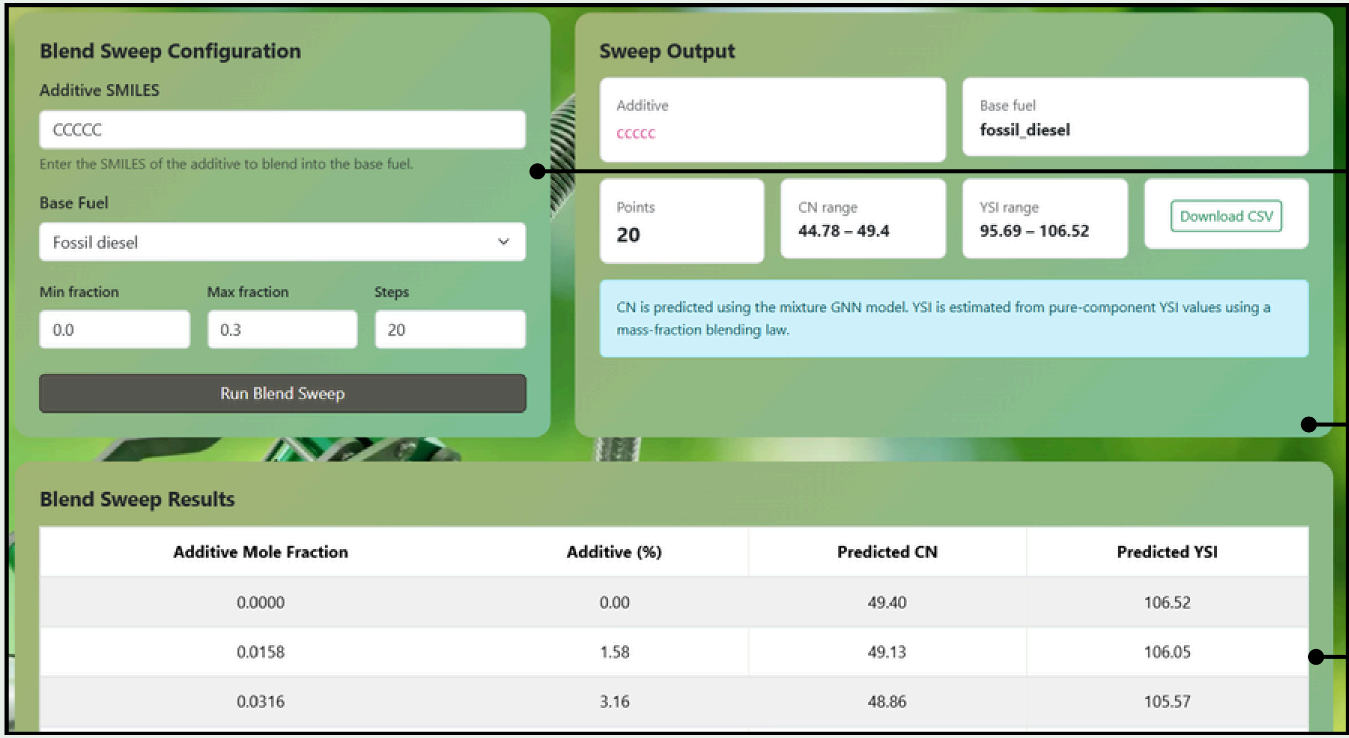
Results include a ranked set of Pareto-optimal candiates for maximum CN and minimum YSI that satisfy fuel property constraints.

Predicted properties for each candidate are also given.

Results can be downloaded as a CSV file

Binary Blending

Select the base fuel from the dropdown menu. The available options are fossil diesel and biodiesel.

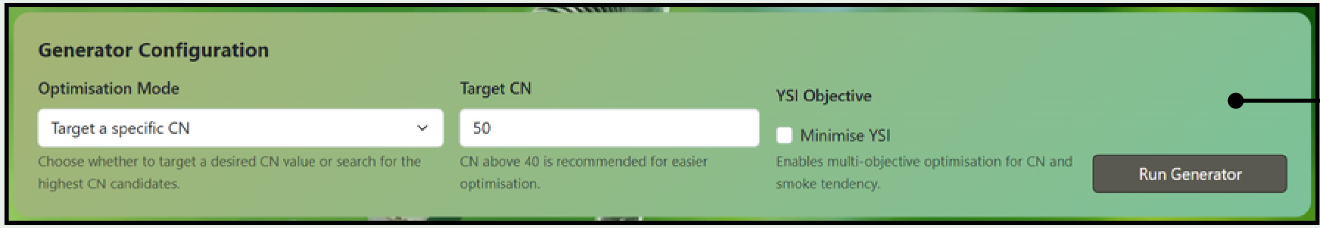


Input
Enter the SMILES of the candidate fuel and adjust the mole fraction range and the steps to customise the sweep

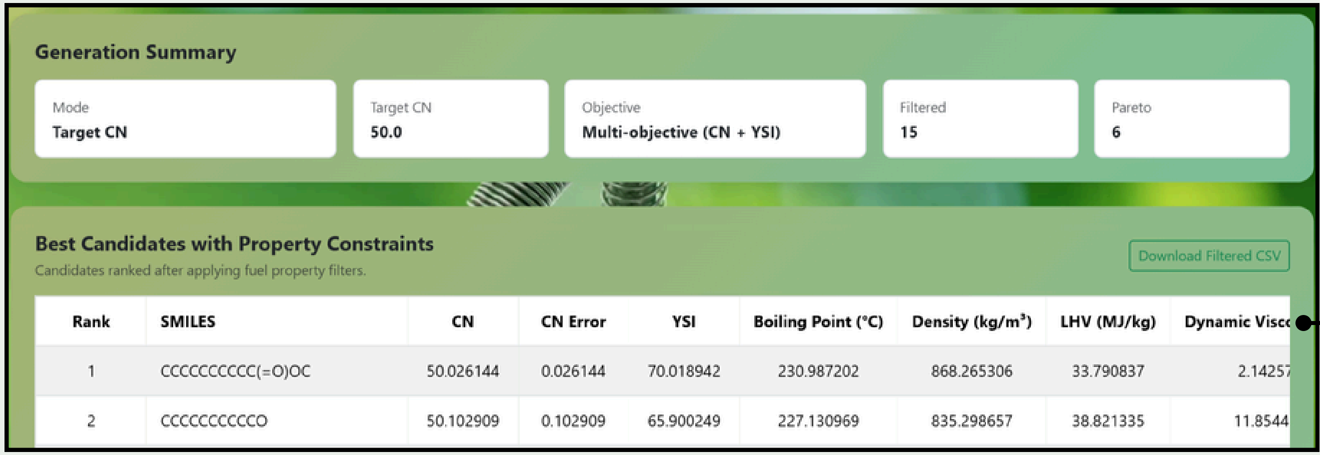
Results
The results include predicted CN and YSI values for each blend. The ranges of estimated values are also given.

Pure Fuel Generator

Select an optimisation mode from the dropdown menu: target a specific CN or maximise CN.



Input
Enter a target CN if running in Target a specific CN mode. Select the option Minimise YSI for multi-objective optimisation.



Rank	SMILES	CN	CN Error	YSI	Boiling Point (°C)	Density (kg/m³)	LHV (MJ/kg)	Dynamic Visc
1	CCCCCCCCCC(=O)OC	50.026144	0.026144	70.018942	230.987202	868.265306	33.790837	2.14257
2	CCCCCCCCCCO	50.102909	0.102909	65.900249	227.130969	835.298657	38.821335	11.8544

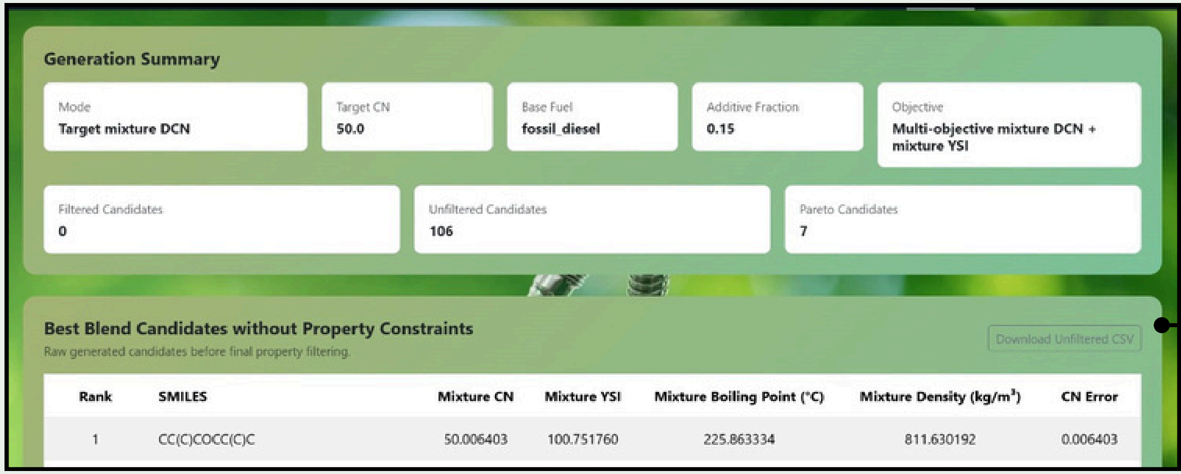
Results
A set of ranked optimal candidates are generated. The results table includes predictions for all six properties.

Results can be downloaded as a CSV file

Blend Component Generator

Select an optimisation mode from the dropdown menu: target a specific CN or maximise CN.

Input
Enter a mole fraction for the blend component and select a base fuel. Enter a target CN if running in <u>Target a specific CN</u> mode and select the option <u>Minimise YSI</u> for multi-objective optimisation.



Results
A set of ranked optimal candidates are generated for the blend component. Results include predicted CN of the resulting blend as well as estimated YSI, boiling point and density.

Results can be downloaded as a CSV file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